

Shrey Patel

Atlanta, Georgia

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Education

Georgia State University

Bachelor of Science in Computer Science – GPA: 3.89, Magna Cum Laude

Aug 2021 – Dec 2024

Atlanta, Georgia

Profile Summary

- Proficient in Python, SQL, and machine learning, with expertise in data science and AI solutions for decision-making and operational optimization.
- Skilled in software engineering principles, including full-stack development, system architecture design, and performance optimization, ensuring scalable and maintainable solutions.
- Experienced in predictive modeling, data analysis, and creating robust AI systems, integrating advanced algorithms with real-world applications.

Relevant Coursework

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|--------------------------------|----------------------------------|---------------------------|------------------------|
| • Fundamentals of Data Science | • Data Mining | • Artificial Intelligence | • Software Engineering |
| • Machine Learning | • Data Structures and Algorithms | • Linear Algebra | • Database Systems |

Experience

Information Technology Intern – Georgia Access, Office of the Commissioner of Insurance and Safety Fire

Atlanta, Georgia | June 2024 – August 2024

- Consolidated SBE-FP briefing, prepared Medicaid Unwinding Reminder Letters, and managed Monthly Recon files for User-Fee using SQL and Power BI to generate ad-hoc and dynamic reports.
- Created applications and cases in enrollment systems, executed and validated FFM test cases, and performed enrollments through EDE partners, comparing the process with the Georgia Access System.
- Compared SSP and SSR artifacts from different years, documented updates, assisted IT Support specialists with laptop and security tasks, and initiated security assessments in MARS-E.

Data Analysis Intern – Pebble Tossers Inc.

Dunwoody, Georgia | February 2024 – August 2024

- Analyzed donation data, membership data, and volunteer hours to identify trends and patterns using Python.
- Utilized SQL and Excel to analyze and visualize data for reporting and decision-making purposes.
- Created interactive visualizations and dashboards using Tableau to effectively communicate donation trends, membership insights, and volunteer hour analytics to stakeholders.

Projects

New Green Approach to Energy-Efficient Neural Network Training

- Developed an energy-efficient neural network training system for air pollution monitoring using SVD and PCA dimensionality reduction techniques, decreasing feature space by 60% (13 to 5 components) while maintaining 92% accuracy, integrated Python/TensorFlow with optimized three-layer architecture (64, 32, 16 units) using ReLU activation and Adam optimizer (learning rate 0.001, batch size 32), reducing CPU utilization from 21.6% to 18.7%, achieving 0.73% energy savings, and decreasing carbon emissions from 342.35 kg to 339.63 kg CO₂ through comprehensive performance evaluations.

Property Rental System

- Built a PHP/MySQL platform for listing, searching, and booking properties, developed backend functionality with custom MySQL schemas, integrated external property data via RapidAPI, while implementing advanced features including customizable rental agreements and automated property availability management

Risk of Alzheimer's: Predictive Modeling for Early Detection

- Led development of an early Alzheimer's detection model at Georgia State University using 373 OASIS Longitudinal dataset records, implementing random forest classifier, support vector machines, and neural networks with k-fold cross-validation, conducted data preprocessing on 12 key clinical features through median imputation of 5% missing data and removal of 15 outliers, achieving 94% prediction accuracy with an 11% reduction in training time through optimized feature selection and hyperparameter tuning, demonstrating potential for clinical applications in early detection.

Technical Skills

- Languages: Python, SQL, Java, C, C++, HTML, CSS, JavaScript
- Frameworks & Libraries: React.js, FastAPI, Node.js, Scikit-Learn, TensorFlow, PyTorch, NumPy, Pandas, Matplotlib, Seaborn
- Databases & Tools: MySQL, MariaDB, RapidAPI, Power BI, Excel, Tableau, Git, XAMPP